FINM3013 Introduction to Financial Derivatives

Chapter 4 Exercises

1. A bank can borrow or lend at LIBOR. Suppose that the six-month rate is 5% and the nine-month rate is 6%. The rate that can be locked in for the period between six months and nine months using an FRA is 7%. What arbitrage opportunities are open to the bank? All rates are continuously compounded.

2. The 6-month, 12-month. 18-month, and 24-month zero rates are 4%, 4.5%, 4.75%, and 5% with semiannual compounding.
   a) What are the rates with continuous compounding?
   b) What is the forward rate for the six-month period beginning in 18 months
   c) What is the value of an FRA that promises to pay you 6% (compounded semiannually) on a principal of $1 million for the six-month period starting in 18 months?

3. The following table gives the prices of bonds

Bond Principal ($)	*Time to Maturity (years)*	*Annual Coupon ($)**	*Bond Price ($)*
100	0.5	0.0	98
100	1.0	0.0	95
100	1.5	6.2	101
100	2.0	8.0	104

*Half the stated coupon is paid every six months

   a) Calculate zero rates for maturities of 6 months, 12 months, 18 months, and 24 months.
   b) What are the forward rates for the periods: 6 months to 12 months, 12 months to 18 months, 18 months to 24 months?
   c) What are the 6-month, 12-month, 18-month, and 24-month par yields for bonds that provide semiannual coupon payments?
   d) Estimate the price and yield of a two-year bond providing a semiannual coupon of 7% per annum.